1 version 1

$$
\min E(Q,T) = \int_{t_1}^{t_F}\int_\Omega p(t)Q(x,t)\,\mathrm{d}x\mathrm{d}t + \frac{\gamma}{2}\int_\Omega |T(x,t_F)-T_{\text{fin}}(x)|^2\mathrm{d}x
$$

$$
\begin{cases} \rho c\dot T(x,t) = \nabla\cdot(k(x)\nabla T(x,t)) + Q(x,t) & \text{in } \Omega\times(0,t_F) \\ -k(x)\partial_\nu T(x,t) = h(x)\left(T(x,t)-T_{\text{out}}(x,t)\right) & \text{on } \partial\Omega\times[0,t_F] \\ T(x,0) = T_{\text{ini}}(x) \end{cases} \tag{SE}
$$

$$
Q\in U_{\text{ad}} = \{Q\in L^2(\Omega\times(0,t_F))\,|\,0\le Q(x,t)\le Q_{\max}(x) \text{ a.e.}\}
$$

$$
\text{optional}:\ T(x,t_F)\ge T_{\text{fin}}(x)(\text{optional if second part of E should be strict})
$$

where $T(x,t)$ temperature in K (Henning: I will use °C, but written K is better), $Q(x,t)$ heating output in $\frac{W}{m^3}$, $p(t)$ price, ρ density in $\frac{kg}{m^3}$, c heat capacity in $\frac{J}{kg\cdot K}$, $k(x)$ heat transfer coefficient in $\frac{W}{m\cdot K}$, $h(x)$ convective heat transfer coefficient in $\frac{W}{m^2K}$, t_1 morning start time

Remember that $\partial_\nu T$ is the outer normal derivative with respect to x and any initial data (or end data) would have to satisfy the PDE.

SE and WSE In order to solve the (SE) we will use a more basic method, in which we seperate between the spacial and temporal derivatives and first solve the PDE in space using approximations of the time derivatives and then the ODE in time or we try to decouple both variables as much as possible to get a linear system in block structure. This is done by Gridap automatically. Thus, we need the weak formulation only in space (so a semi-weak formulation of the whole system). Here we will give it in its weak form remarking that every time integral of the (WSE) for the (SWF) would be replaced by "for all $t\in(0,t_F)$", because computing the derivative of the energy functional is easier this way without loosing track of the t, and get with the test space $H^1_\Gamma(\Omega\times(0,t_F))$ where $\Gamma=\Omega\times\{0\}$

$$
\begin{aligned}
\int_0^{t_F}\int_\Omega \rho c\dot T(x,t)\phi(x,t)\,\mathrm{d}x\mathrm{d}t &= \int_0^{t_F}\int_\Omega \nabla\cdot(k(x)\nabla T(x,t))\phi(x,t) + Q(x,t)\phi(x,t)\,\mathrm{d}x\mathrm{d}t \\
&= \int_0^{t_F}\int_\Omega -k(x)\nabla T(x,t)\cdot\nabla\phi(x,t) + Q(x,t)\phi(x,t)\,\mathrm{d}x\mathrm{d}t + \int_0^{t_F}\int_{\partial\Omega} k(x)\partial_\nu T(y,t)\phi(x,t)\,\mathrm{d}y\mathrm{d}t \\
&= \int_0^{t_F}\int_\Omega -k(x)\nabla T(x,t)\cdot\nabla\phi(x,t) + Q(x,t)\phi(x,t)\,\mathrm{d}x\mathrm{d}t - \int_0^{t_F}\int_{\partial\Omega} h(y)(T(y,t)-T_{\text{out}}(y,t))\phi(x,t)\,\mathrm{d}y\mathrm{d}t
\end{aligned}
$$

giving us the weak formulation

$$
\int_0^{t_F}\int_\Omega \rho c\dot T(x,t)\phi(x,t)\,\mathrm{d}x\mathrm{d}t = \int_0^{t_F}\int_\Omega -k(x)\nabla T(x,t)\cdot\nabla\phi(x,t) + Q(x,t)\phi(x,t)\,\mathrm{d}x\mathrm{d}t - \int_0^{t_F}\int_{\partial\Omega} h(y)\left(T(y,t)-T_{\text{out}}(y)\right)\phi(y,t)\,\mathrm{d}y\mathrm{d}t \tag{WSE}
$$

and thus

$$
\begin{aligned}
a_{\text{SE}}(T,\phi) &= \int_0^{t_F}\int_\Omega \rho c\dot T(x,t)\phi(x,t) + k(x)\nabla T(x,t)\cdot\nabla\phi(x,t)\,\mathrm{d}x\mathrm{d}t + \int_0^{t_F}\int_{\partial\Omega} h(y)T(y,t)\phi(y,t)\,\mathrm{d}y\mathrm{d}t, \\
l_{\text{SE}}(\phi) &= \int_0^{t_F}\int_\Omega Q(x,t)\phi(x,t)\,\mathrm{d}x\mathrm{d}t + \int_0^{t_F}\int_{\partial\Omega} h(y)T_{\text{out}}(y)\phi(y,t)\,\mathrm{d}y\mathrm{d}t
\end{aligned}
$$

By Lax-Milgram there exists a unique solution $T(x,t)$ and thus we can define a solution operator $S: L^2(\Omega\times(0,t_F))\to H^1(\Omega\times(0,t_F))$, $S(Q)=T$.

Henning: remember that each boundary integral could be rewritten with the Trace operator, might be helpful for the next part. When writing the (WSE) the open interval for $t\in(0,t_F)$ is important.

Lagrangian + AE From the optimisation functional and the (WSE) we get the Lagrangian as

$$L(T,Q,W)=\int_{t_1}^{t_F}\int_\Omega p(t)Q(x,t)\,\mathrm{d}x\mathrm{d}t+\frac{\gamma}{2}\int_\Omega |T(x,t_F)-T_{\text{fin}}(x)|^2\mathrm{d}x+\int_0^{t_F}\int_\Omega \rho c\dot{T}(x,t)W(x,t)+k(x)\nabla T(x,t)\cdot\nabla W(x,t)-Q(x,t)W(x,t)\,\mathrm{d}x\mathrm{d}t+\int_0^{t_F}\int_{\partial\Omega} h(y)\left(T(y,t)-T_{\text{out}}(y)\right)W(y,t)\,\mathrm{d}y\mathrm{d}t$$

which gives us the weak formulation of the adjoint (AE) as (notice that the second term contains $\dot{\psi}$ which in the (SWF) would be zero and thus needs to be solved differently: in that case we are actually only looking for the derivative of the x dendent T, so that the derivative $\partial_{T(x)}T(x,t)$ does not give 1 but a time dependent function)

$$L_T(T,Q,W)\psi=\gamma\int_\Omega\left(T(x,t_F)-T_{\text{fin}}(x)\right)\psi(x,t_F)\mathrm{d}x+\int_0^{t_F}\int_\Omega \rho c\dot{\psi}(x,t)W(x,t)+k(x)\nabla\psi(x,t)\cdot\nabla W(x,t)\,\mathrm{d}x\mathrm{d}t+\int_0^{t_F}\int_{\partial\Omega} h(y)\psi(y,t)W(y,t)\,\mathrm{d}y\mathrm{d}t$$

By Green for any $t\in(0,t_F)$ it is

$$\int_\Omega k(x)\nabla\psi(x,t)\cdot\nabla W(x,t)\,\mathrm{d}x=-\int_\Omega\nabla\cdot\left(k(x)\nabla W(x,t)\right)\psi(x,t)\,\mathrm{d}x+\int_{\partial\Omega}k(y)\partial_\nu W(y,t)\psi(y,t)\,\mathrm{d}y$$

and from Green for the t variable (integration by parts) we get

$$\int_0^{t_F}\int_\Omega \rho c\dot{\psi}(x,t)W(x,t)\,\mathrm{d}x\mathrm{d}t=-\int_0^{t_F}\int_\Omega \rho c\psi(x,t)\dot{W}(x,t)\,\mathrm{d}x\mathrm{d}t+\left[\int_\Omega \rho c\psi(x,t)W(x,t)\,\mathrm{d}x\right]_0^{t_F}=-\int_0^{t_F}\int_\Omega \rho c\psi(x,t)\dot{W}(x,t)\,\mathrm{d}x\mathrm{d}t+\int_\Omega \rho c\psi(x,t_F)W(x,t_F)\,\mathrm{d}x$$

when assuming $W(x,0)=0$ a.e. (because W has to be able to operate as test function for (SE)) and thus $\psi\in H^1_\Gamma(\Omega\times(0,t_F))$. With these it is

$$L_T(T,Q,W)\psi=\gamma\int_\Omega\left(T(x,t_F)-T_{\text{fin}}(x)\right)\psi(x,t)\mathrm{d}x-\int_0^{t_F}\int_\Omega \rho c\psi(x,t)\dot{W}(x,t)+\nabla\cdot\left(k(x)\nabla W(x,t)\right)\psi(x,t)\,\mathrm{d}x\mathrm{d}t-\int_0^{t_F}\int_{\partial\Omega} h(y)\psi(y,t)W(y,t)-k(y)\partial_\nu W(y,t)\psi(y,t)\,\mathrm{d}y\mathrm{d}t+\int_\Omega \rho c\psi(x,t_F)W(x,t_F)\,\mathrm{d}x$$

giving the strong version

$$\begin{cases}\rho c\dot{W}(x,t)=-\nabla\cdot\left(k(x)\nabla W(x,t)\right) & \text{in }\Omega\times(0,t_F)\\ k(y)\partial_\nu W(y,t)=h(y)W(y,t) & \text{on }\partial\Omega\times[0,t_F]\\ W(x,t_F)=-\dfrac{\gamma}{\rho c}\left(T(x,t_F)-T_{\text{fin}}(x)\right) & \text{in }\Omega\end{cases}\tag{AE}$$

Thus, we can see that the (AE) has to be solved backwards in time. The weak version is

$$\int_0^{t_F}\int_\Omega \rho c\psi(x,t)\dot{W}(x,t)-k(x)\nabla\psi(x,t)\cdot\nabla W(x,t)\,\mathrm{d}x\mathrm{d}t=-\int_0^{t_F}\int_{\partial\Omega} h(y)\psi(y,t)W(y,t)\,\mathrm{d}y\mathrm{d}t\tag{WAE}$$

For the (AE) we have

$$a_{\text{AE}}(W,\psi)=\int_0^{t_F}\int_\Omega \rho c\psi(x,t)\dot{W}(x,t)-k(x)\nabla\psi(x,t)\cdot\nabla W(x,t)\,\mathrm{d}x\mathrm{d}t+\int_0^{t_F}\int_{\partial\Omega} h(y)\psi(y,t)W(y,t)\,\mathrm{d}y\mathrm{d}t\qquad l_{\text{AE}}(\psi)=0$$

By Lax-Milgram there exists a unique solution $W(x,t)$ and thus a solution operator $S^*:L^2(\Omega)\to H^1(\Omega\times(0,t_F))$, $S^*(T(x,t_F)-T_{\text{fin}}(x))=W(x,t)$.

Henning: Actually, we get the (AE) by the variation principle and not by differentiation, which does not fully make sense here (see 1.1).

adjoint forward Since the (AE) has to be solved backwards in time, which is not possible with the software we are using (although implementation would be just as is) we introduce $\widetilde{W}(x,t) = W(x, t_F - t)$. By the (AE) this function satisfies

$$\begin{cases} \rho c\dot{\widetilde{W}}(x,t) = \nabla \cdot \left(k(x)\nabla\widetilde{W}(x,t)\right) & \text{in } \Omega \times (0,t_F) \\ k(y)\partial_\nu \widetilde{W}(y,t) = h(y)\widetilde{W}(y,t) & \text{on } \partial\Omega \times [0,t_F] \\ \widetilde{W}(x,0) = -\frac{\gamma}{\rho c}\left(T(x,t_F) - T_{\text{fin}}(x)\right) & \text{in } \Omega \end{cases} \tag{AE'}$$

which can now be solved forward in time. The weak version is

$$\int_0^{t_F}\int_\Omega \rho c\psi(x,t)\dot{\widetilde{W}}(x,t) + k(x)\nabla\psi(x,t)\cdot\nabla\widetilde{W}(x,t)\,\mathrm{d}x\mathrm{d}t = \int_0^{t_F}\int_{\partial\Omega} h(y)\psi(y,t)\widetilde{W}(y,t)\,\mathrm{d}y\mathrm{d}t \tag{WAE}$$

For the (AE') we have

$$a_{\text{AE}}(\widetilde{W},\psi) = \int_0^{t_F}\int_\Omega \rho c\psi(x,t)\dot{\widetilde{W}}(x,t) + k(x)\nabla\psi(x,t)\cdot\nabla\widetilde{W}(x,t)\,\mathrm{d}x\mathrm{d}t - \int_0^{t_F}\int_{\partial\Omega} h(y)\psi(y,t)\widetilde{W}(y,t)\,\mathrm{d}y\mathrm{d}t \qquad l_{\text{AE}}(\psi) = 0 \tag{1}$$

which then gives us the original adjoint as $W(x,t) = \widetilde{W}(x, t_F - t)$.

1.1 Variation

In order to find the adjoint, we introduce a small variation δT of T in the respective space and look at the difference this variation makes on L (this is like computing $L(T_1, Q, W) - L(T_2, Q, W)$, remember that the adjoint is a measure for the sensitivity of the functional with respect to T).

$$\begin{aligned} \delta_T L(T,Q,W) &= \frac{\gamma}{2}\int_\Omega \delta_T|T(x,t_F) - T_{\text{fin}}(x)|^2\mathrm{d}x + \int_0^{t_F}\int_\Omega \rho c\dot{\delta T}(x,t)W(x,t) - k(x)\nabla\delta T(x,t)\cdot\nabla W(x,t)\,\mathrm{d}x\mathrm{d}t - \int_0^{t_F}\int_{\partial\Omega} h(y)\left(\delta T(y,t) - T_{\text{out}}(y)\right)W(y,t)\,\mathrm{d}y\mathrm{d}t \\ &= \gamma\int_\Omega (T(x,t_F) - T_{\text{fin}}(x))\delta T(x,t_F)\mathrm{d}x - \int_0^{t_F}\int_\Omega \rho c\delta T(x,t)\dot{W}(x,t)\,\mathrm{d}x\mathrm{d}t + \int_\Omega \rho c\delta T(x,t_F)W(x,t_F)\,\mathrm{d}x - \int_0^{t_F}\int_\Omega k(x)\nabla\delta T(x,t)\cdot\nabla W(x,t)\,\mathrm{d}x\mathrm{d}t - \int_0^{t_F}\int_{\partial\Omega} h(y)\left(\delta T(y,t) - T_{\text{out}}(y)\right)W(y,t)\,\mathrm{d}y\mathrm{d}t \end{aligned}$$

where we used the same as above and $\delta T(x,0) = 0$ because of the initial value condition, which fixes the Ts there and because the term is linear. Also for the first term we used Taylor giving us $x^2 = x_0^2 + 2x(x - x_0) + o((x-x_0)^2)$, giving us $\delta_x(x-a)^2 = (x-a)^2 - (x_0-a)^2 = 2(x-a)(x-x_0) + o(x-x_0) = 2(x-a)\delta_x$.

Now we introduce the solution operator S, which for a given Q gives the solution T of the (SE), so $T = S(Q)$. With this we can introduce the reduced energy functional

$$e(Q) = \int_{t_1}^{t_F}\int_\Omega p(t)Q(x,t)\,\mathrm{d}x\mathrm{d}t + \frac{\gamma}{2}\int_\Omega |S(Q)(x,t_F) - T_{\text{fin}}(x)|^2\mathrm{d}x \tag{2}$$

In order to find the gradient of e we differentiate it and find by noting that S is linear and therefore has derivative $S'(Q)h = Sh$

$$\begin{aligned} e'(Q)h &= \int_{t_1}^{t_F}\int_\Omega p(t)h(x,t)\,\mathrm{d}x\mathrm{d}t + \gamma\int_\Omega (S(Q)(x,t_F) - T_{\text{fin}}(x))Sh\mathrm{d}x & (3) \\ &= \langle p|h\rangle + \langle S(Q) - T_{\text{fin}}|Sh\rangle|_{t=t_F} = \langle p|h\rangle + \gamma\langle S^*(S(Q) - T_{\text{fin}})|h\rangle|_{t=t_F} \equiv p - \rho cW & (4) \end{aligned}$$

where S^* is the solution operator for (AE) and W is the solution ....

2 version 2

$$\begin{aligned}
&\min E(Q,T) = \int_{t_1}^{t_F}\int_\Omega (p(t)Q(x,t))^2\,\mathrm{d}x\mathrm{d}t + \frac{\gamma}{2}\int_\Omega |T(x,t_F)-T_{\text{fin}}(x)|^2\mathrm{d}x \\
&\begin{cases} \rho c\dot{T}(x,t) = \nabla\cdot(k(x)\nabla T(x,t)) + Q(x,t) & \text{in } \Omega\times(0,t_F) \\ -k(x)\partial_\nu T(x,t) = h(x)\,(T(x,t)-T_{\text{out}}(x,t)) & \text{on } \partial\Omega\times[0,t_F] \\ T(x,0) = T_{\text{ini}}(x) \end{cases} \\
&Q\in U_{\text{ad}} = \{Q\in L^2(\Omega\times(0,t_F))\,|\,0\le Q(x,t)\le Q_{\max}(x)\text{ a.e.}\}
\end{aligned} \tag{SE}$$

In this case everything stays the same but the Lagrangian is

$$L(T,Q,W) = \int_{t_1}^{t_F}\int_\Omega (p(t)Q(x,t))^2\,\mathrm{d}x\mathrm{d}t + \frac{\gamma}{2}\int_\Omega |T(x,t_F)-T_{\text{fin}}(x)|^2\mathrm{d}x + \int_0^{t_F}\int_\Omega \rho c\dot{T}(x,t)W(x,t) - k(x)\nabla T(x,t)\cdot\nabla W(x,t) - Q(x,t)W(x,t)\,\mathrm{d}x\mathrm{d}t - \int_0^{t_F}\int_{\partial\Omega} h(y)\,(T(y,t)-T_{\text{out}}(y))\,W(y,t)\,\mathrm{d}y\mathrm{d}t \tag{5}$$

Solution operator + gradient Now we introduce the solution operator S, which for a given Q gives the solution T of the (SE), so $T=S(Q)$. With this we can introduce the reduced energy functional

$$e(Q) = \int_{t_1}^{t_F}\int_\Omega (p(t)Q(x,t))^2\,\mathrm{d}x\mathrm{d}t + \frac{\gamma}{2}\int_\Omega |S(Q)(x,t_F)-T_{\text{fin}}(x)|^2\mathrm{d}x \tag{6}$$

In order to find the gradient of e we differentiate it and find by noting that S is linear and therefore has derivative $S'(Q)h = Sh$

$$\begin{aligned}
e'(Q)h &= \int_{t_1}^{t_F}\int_\Omega 2p^2(t)Q(x,t)h(x,t)\,\mathrm{d}x\mathrm{d}t + \gamma\int_\Omega (S(Q)(x,t_F)-T_{\text{fin}}(x))Sh\mathrm{d}x \\
&= \langle 2p^2Q|h\rangle + \langle S(Q)-T_{\text{fin}}|Sh\rangle|_{t=t_F} = 2\langle p^2Q|h\rangle + \langle S^*(S(Q)-T_{\text{fin}})|h\rangle|_{t=t_F} \equiv 2p^2Q - W
\end{aligned}$$

Thus, without box constraints from the condition $e'(Q)=0$ we would get $Q = -\frac{1}{2p^2}W$ and can use the projection of that onto the box for our case (see bang bang).

We can also get the gradient by the (VI) which we get by $L_Q = 0$ as

$$0 = L_Q(T,Q,W)h = 2\int_{t_1}^{t_F}\int_\Omega p(t)Q(x,t)h(x,t)\,\mathrm{d}x\mathrm{d}t - \int_0^{t_F}\int_\Omega W(x,t)h(x,t)\,\mathrm{d}x\mathrm{d}t = \langle 2p^2Q - W, h\rangle_{L^2(\Omega\times(0,t_F))}$$

This property has to hold for every $h\in Q_{\text{ad}} - Q$ for the Q from the integral, which gives us the bang-bang result.

Optimal step size We found out that the gradient of the reduced cost is $\nabla e(Q) = 2p^2Q - \rho cW$, so in the gradient descent algorithm we will go into the negative of that direction and we can actually compute the optimal step size. Consider a given iteration k and the computed control Q_k, state T_k and adjoint W_k. Also let $v_k = -\nabla e(Q) = -(2p^2Q - \rho cW)$, $I = [t_0,t_F]$ and define as the step-length-dependent function we want to minimise in the iteration

$$g(s) = e(Q_k + sv_k)$$

Then we can rewrite

$$\begin{aligned}
g(s) &= e(Q_k+sv_k) \\
&= \|p(Q_k+sv_k)\|^2_{L^2(\Omega\times I)} + \frac{\gamma}{2}\|S(Q_k+sv_k) - T_{\text{fin}}\|^2_{L^2(\Omega)}\big|_{t=t_F} \\
&= \|pQ_k\|^2_{L^2(\Omega\times I)} + 2s\langle p^2Q_k, v_k\rangle_{L^2(\Omega\times I)} + s^2\|pv_k\|^2_{L^2(\Omega\times I)} + \frac{\gamma}{2}\|T_k - T_{\text{fin}}\|^2_{L^2(\Omega)}\big|_{t=t_F} + \gamma s\langle T_k - T_{\text{fin}}, Sv_k\rangle_{L^2(\Omega)}\big|_{t=t_F} + s^2\frac{\gamma}{2}\|Sv_k\|^2_{L^2(\Omega)}\big|_{t=t_F} \\
&= s^2\Big(\|pv_k\|^2_{L^2(\Omega\times I)} + \frac{\gamma}{2}\|Sv_k\|^2_{L^2(\Omega)}\big|_{t=t_F}\Big) + s\Big(2\langle p^2Q_k, v_k\rangle_{L^2(\Omega\times I)} + \gamma\langle T_k - T_{\text{fin}}, Sv_k\rangle_{L^2(\Omega)}\big|_{t=t_F}\Big) + 1\Big(\|pQ_k\|^2_{L^2(\Omega\times I)} + \frac{\gamma}{2}\|T_k - T_{\text{fin}}\|^2_{L^2(\Omega)}\big|_{t=t_F}\Big) \\
&= s^2\Big(\|pv_k\|^2_{L^2(\Omega\times I)} + \frac{\gamma}{2}\|Sv_k\|^2_{L^2(\Omega)}\big|_{t=t_F}\Big) + s\Big(2\langle p^2Q_k, v_k\rangle_{L^2(\Omega\times I)} + \gamma\langle S^*(T_k - T_{\text{fin}}), v_k\rangle_{L^2(\Omega)}\big|_{t=t_F}\Big) + 1\Big(\|pQ_k\|^2_{L^2(\Omega\times I)} + \frac{\gamma}{2}\|T_k - T_{\text{fin}}\|^2_{L^2(\Omega)}\big|_{t=t_F}\Big) \\
&= s^2\Big(\|pv_k\|^2_{L^2(\Omega\times I)} + \frac{\gamma}{2}\|Sv_k\|^2_{L^2(\Omega)}\big|_{t=t_F}\Big) + s\Big(2\langle p^2Q_k, v_k\rangle_{L^2(\Omega\times I)} + \langle \rho cW_k, v_k\rangle_{L^2(\Omega)}\big|_{t=t_F}\Big) + 1\Big(\|pQ_k\|^2_{L^2(\Omega\times I)} + \frac{\gamma}{2}\|T_k - T_{\text{fin}}\|^2_{L^2(\Omega)}\big|_{t=t_F}\Big) \\
&= s^2\Big(\|pv_k\|^2_{L^2(\Omega\times I)} + \frac{\gamma}{2}\|Sv_k\|^2_{L^2(\Omega)}\big|_{t=t_F}\Big) - s\|v_k\|^2_{L^2(\Omega\times I)} + 1\Big(\|pQ_k\|^2_{L^2(\Omega\times I)} + \frac{\gamma}{2}\|T_k - T_{\text{fin}}\|^2_{L^2(\Omega)}\big|_{t=t_F}\Big)
\end{aligned}$$

This is a quadratic function in s and has the minimum in

$$s_{\min} = \frac{\|v_k\|^2_{L^2(\Omega\times I)}}{2\left(\|pv_k\|^2_{L^2(\Omega\times I)} + \frac{\gamma}{2}\|Sv_k\|^2_{L^2(\Omega)}\big|_{t=t_F}\right)}$$

which requires to solve the (SE) again with input v_k instead of Q_k.

3 Version 3

$$\min E(Q,T) = \int_{t_1}^{t_F}\int_\Omega p(t)Q(x,t)\,\mathrm{d}x\mathrm{d}t$$

$$\begin{cases} \rho c\dot{T}(x,t) = \nabla\cdot(k(x)\nabla T(x,t)) + Q(x,t) & \text{in } \Omega\times(0,t_F) \\ -k(x)\partial_\nu T(x,t) = h(x)\left(T(x,t) - T_{\text{out}}(x,t)\right) & \text{on } \partial\Omega\times[0,t_F] \\ T(x,0) = T_{\text{ini}}(x) \end{cases} \tag{SE}$$

$$Q\in U_{\text{ad}} = \{Q\in L^2(\Omega\times(0,t_F))\,|\,0\le Q(x,t)\le Q_{\max}(x)\text{ a.e.}\}$$

$$T(x,t_F)\ge T_{\text{fin}}(x)$$

The (SE) and thus (WSE) is the same. But we get a new Lagrangian multiplier $\mu\ge 0$ and the Lagrangian is

$$L(T,q,W,\mu) = \int\limits_{t_1}^{t_F}\int\limits_\Omega p(t)Q(x,t)\,\mathrm{d}x\mathrm{d}t + \int\limits_0^{t_F}\int\limits_\Omega \rho c\dot{T}(x,t)W(x,t) + k(x)\nabla T(x,t)\cdot\nabla W(x,t) - Q(x,t)W(x,t)\,\mathrm{d}x\mathrm{d}t + \int\limits_0^{t_F}\int\limits_{\partial\Omega} h(y)\left(T(y,t)-T_{\text{out}}(y)\right)W(y,t)\,\mathrm{d}y\mathrm{d}t + \int\limits_\Omega \mu(x)(T(x,t_F) - T_{\text{fin}}(x))\mathrm{d}x$$

4 Projection Property

Theorem 1. *Let V be a linear space (=vector space), $M\subseteq V$ a closed convex set, $f: V\to\mathbb{R}$ a strictly convex function and consider the unconstraint and constraint minimisation problems*

$$\min_{x\in V} f(x) \qquad \text{and} \qquad \min_{x\in M} f(x).$$

Then each problem has at most one solution and if the unconstrained problem has a solution $x_V\in\operatorname{argmin}_{x\in V} f(x)$ then the projection $\pi(x_V)$ is the solution to the constrained problem.

Proof. (i) Since M is arbitrary, proofing that the constrained problem has at most one solution proofs the claim also for the other case by setting $M = V$. Assuming there are two solution $x^1, x^2\in M$, then $(x^1,x^2)\subseteq M$ by convexity of M and for every $x\in(x^1,x^2)$ by strict convexity of f it is $f(x) < \lambda f(x^1) + (1-\lambda)f(x^2) = 1\cdot\min\limits_{y\in M} f(y)$, contradicting the minimality of x^1, x^2.

(ii) Assume the minimum over M, denoted by x_M, would not be the projection. Then $f(x_V) < f(\pi(x_V))$, $f(x_V) < f(x_M)$ by (i) and $f(x_M) < f(\pi(x_V))$ and thus $f(x_V) < f(x_M) < f(\pi(x_V))$. x_M has to be in the boundary of M, otherwise we could as in (i) go a little bit into direction of x_V (better: the direction is $x_V - x_M$) and have smaller vallues. Connecting any two points on the trianle $T = \operatorname{span}\{x_V, x_M, \pi(x_V)\}$ we thus get $f(x) < f(\pi(x_V))$ for every $x\in T\setminus\{\pi(x_V)\}$ □

5 Remark: minimal time interval

If we consider the system without the boundary term, so without heat loss to the outside, then in order to be able to achieve an end temperature close to T_{fin} we need enough time, precisely

$$t = \frac{|\Omega|}{|Q_{\text{ad}}|}\frac{T_{\text{fin}} - T_{\text{ini}}}{\max Q}$$

much time, which corresponds to heating the room in average to the final temperature using maximum output all the time.

For small time intervalls we either get a bigger error in the corresponding term of the energy or need a big $\max Q$. Looking at the Theta-method with Y_n being any solution (of SE or AE) for only a given time t_n the next solution for t_{n+1} with the help of the right hand side extracted by looking at the time change of the system will be

$$Y_{n+1} = Y_n + \Delta t\left(\theta F(t_n, Y_n) + (1-\theta)F(t_{n+1}, Y_{n+1})\right)$$

which gets solved by the Newton method iterating over k

$$Y_{n+1}^{k+1} = Y_{n+1}^{k} - \left(G'(Y_{n+1}^{k})\right)^{-1} G(Y_{n+1}^{k})$$

where

$$G(Y) = Y - \Delta t\left(\theta F(t_n, Y_n) + (1-\theta)F(t+h, Y)\right) - Y_n, \qquad G'(Y) = I - (1-\theta)\Delta t \partial_Y F(t_{n+1}, Y)$$

6 Adjoint

Theorem 2. *The Adjoint solution operator from version 2 satisfies*

$$\langle S^*(T - T_{fin}), h\rangle = \langle T - T_{fin}, S(h)\rangle$$

Proof. We remind about the weak forms (WSE) with test function W and (WAE) (from L_T before using Green) with test function T

$$\int_0^{t_F}\int_\Omega \rho c\dot{T}(x,t)W(x,t)\,\mathrm{d}x\mathrm{d}t = \int_0^{t_F}\int_\Omega -k(x)\nabla T(x,t)\cdot\nabla W(x,t) + Q(x,t)W(x,t)\,\mathrm{d}x\mathrm{d}t - \int_0^{t_F}\int_{\partial\Omega} h(y)\left(T(y,t) - T_{\mathrm{out}}(y)\right)W(y,t)\,\mathrm{d}y\mathrm{d}t \tag{WSE}$$

$$\int_0^{t_F}\int_\Omega \rho c\dot{T}(x,t)W(x,t)\,\mathrm{d}x\mathrm{d}t = -\int_0^{t_F}\int_\Omega k(x)\nabla T(x,t)\cdot\nabla W(x,t)\,\mathrm{d}x\mathrm{d}t - \gamma\int_\Omega \left(T(x,t_F) - T_{\mathrm{fin}}(x)\right)T(x,t_F)\mathrm{d}x - \int_0^{t_F}\int_{\partial\Omega} h(y)W(y,t)T(y,t)\,\mathrm{d}y\mathrm{d}t \tag{WAE}$$

Subtracting (WAE) from (WSE) we get

$$0 = \int_0^{t_F}\int_\Omega Q(x,t)W(x,t)\,\mathrm{d}x\mathrm{d}t + \int_0^{t_F}\int_{\partial\Omega} h(y)T_{\mathrm{out}}(y)W(y,t)\,\mathrm{d}y\mathrm{d}t + \gamma\int_\Omega \left(T(x,t_F) - T_{\mathrm{fin}}(x)\right)T(x,t_F)\mathrm{d}x$$

With these it is

$$\begin{aligned}
\langle W, S(Q)\rangle &= \int_0^{t_F}\int_\Omega W(x,t)T(x,t)\,\mathrm{d}x\mathrm{d}t \\
&= \int_0^{t_F}\int_\Omega W(x,t)\left(T_{\mathrm{ini}}(x) + \int_0^t \dot{T}(x,s)\,\mathrm{d}s\right)\mathrm{d}x\mathrm{d}t \\
&= \int_0^{t_F}\int_\Omega W(x,t)T_{\mathrm{ini}}(x) + \int_0^t \dot{T}(x,s)W(x,t)\,\mathrm{d}s\,\mathrm{d}x\mathrm{d}t \\
&= \int_0^{t_F}\int_\Omega W(x,t)T_{\mathrm{ini}}(x) + \int_0^t \dot{T}(x,s)W(x,t)\,\mathrm{d}s\,\mathrm{d}x\mathrm{d}t \\
&=
\end{aligned}$$

□

In general it looks like this

Theorem 3. *Assume the problem is*

$$\min E(Q,T) \quad \textit{such that} \quad N(Q,T) = 0$$

where N contains only linear Terms of Q and T.

Proof. The Lagrangian is

$$L(Q,T,W) = E(Q,T) + WN(Q,T)$$

and the adjoint equation is

$$L_T(Q,T,W)h = E_T(Q,T) + WN_T(Q,T)h$$

□

7 Pontryagin's minimum (maximum) principle

See [1]. We can also get the optimal adjoint by Pontryagin's minimum principle, which may be used here to even out some holes in our argumentation. We assume version 2.

Theorem 4 (Pontryagin's minimum principle (version [1]))**.** *Consider the problem*

$$\min_{q\in Q} \Phi(y(t_f)) + \int_0^{t_f} F\big(y(t), q(t)\big)\,\mathrm{d}t$$
$$\dot{y} = f\big(y(t), q(t)\big), \quad y(0) = y_0$$

Let the Hamiltonian be given by $H(q,y,p,t) = F\big(y(t),q(t)\big) + p(t)f\big(y(t),q(t)\big)$*. Then the optimal adjoint solves*

$$\dot{p} = H_y \qquad p(t_f) = \Phi_y(y(t_f))$$

Identifying the functions for our problem and using $y = T$, $q = Q$, $p = W$ we get (notice that here we write everything in the strong version)

$$\begin{aligned}
\Phi(T(t_f)) &= \int_\Omega \left(T(x,t_f) - T_{\text{fin}}(x)\right)^2 \mathrm{d}x \\
F\big(T(t), Q(t)\big) &= \int_\Omega (pQ)^2\,\mathrm{d}x \\
f\big(T(t), Q(t)\big) &= \frac{1}{\rho c}\left(\nabla \cdot (k(x)\nabla T(x,t)) + Q(x,t)\right) \\
H(Q,T,W,t) &= \int_\Omega (pQ)^2\,\mathrm{d}x - \int_0^{t_F}\int_\Omega k(x)\nabla T(x,t)\cdot\nabla W(x,t) - Q(x,t)W(x,t)\,\mathrm{d}x\mathrm{d}t - \int_0^{t_F}\int_{\partial\Omega} h(y)\left(T(y,t) - T_{\text{out}}(y)\right)W(y,t)\,\mathrm{d}y\mathrm{d}t
\end{aligned}$$

and thus the adjoint satisfies

$$\dot{W}(x,t)h =$$

8 Other

- Remember the Courant-Friedrichs-Lewy (CFL) condition: If the numerical solution is screwed up it might be because space step length / time step length is too small. Honestly, I do not know the specific condition for that problem, it might even be reversed here, but this should be considered.
- Galerkin method or Ritz method - mention them and I think we use Galerkin
- Pontryagin's maximum principle??? Gives the adjoint of a system like ours, but it might be hard to transform

References

[1] Peter Jaksch. "Implementation of an adjoint thermal solver for inverse problems". In: *European Conference on Turbomachinery Fluid Dynamics and hermodynamics*. Stockholm, Sweden, 2017. DOI: `10.29008/ETC2017-336`. URL: `http://www.euroturbo.eu/publications/proceedings-papers/ETC2017-336/` (visited on 04/06/2025).